

Multiple Choice

1. **Answer:** D

Options: A) By observing the cell under a microscope; B) Using fluorescent tagging of proteins; C) Tracking changes in cell size and shape; D) Autoradiography with labelled amino acids; E) Through protein synthesis

Note: Correct option: D - Autoradiography with labelled amino acids

2. **Answer:** D

Options: A) Flash flood; B) Tsunami; C) Sudden reduction in the animal food resource; D) Destruction of the ozone layer; E) Drought

Note: Correct option: D - Destruction of the ozone layer

3. **Answer:** A

Options: A) Oncogenes are found only in viruses; B) Oncogenes prevent cancer cells; C) Oncogenes are a type of immune cell; D) Oncogenes detoxify harmful substances in the cell; E) Oncogenes slow down cell division

Note: Correct option: A - Oncogenes are found only in viruses

4. **Answer:** C

Options: A) blastoderm; B) mesoderm; C) amnion; D) zona pellucida; E) ectoderm

Note: Correct option: C - amnion

5. **Answer:** B

Options: A) Savanna; B) Taiga; C) Tundra; D) Chaparral; E) Alpine

Note: Correct option: B - Taiga

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Present audit shows that the process of implementation of labor analgesia was quick, successful and safe, notwithstanding the identification of one cluster of women with suboptimal response to epidural analgesia that need to be further studies, overall pregnant womens'adhesion to labor analgesia was satisfactory.

7. **Answer:** True

Options: A) True; B) False

Note: Patients undergoing cardiac surgery who are exposed to intraoperative precursor events were more likely to experience a postoperative MACE. Quality improvement techniques aimed at mitigating the consequences of precursor events might improve the surgical outcomes for cardiac surgical patients.

8. **Answer:** True

Options: A) True; B) False

Note: Our results suggest that inhaled steroids are better than cromones in preventing admissions for asthma when two provinces with different practices for maintenance medication of steady-state asthma were compared.

9. **Answer:** True

Options: A) True; B) False

Note: Although the exact mechanism of SSDH in this case is unclear, we speculate that this SSDH was a hematoma that migrated from the intracranial subdural space. Low CSF pressure because of continuous drainage and intrathecal thrombolytic therapy may have played an important role in the migration of the hematoma through the spinal canal. It is important to recognize the SSDH as a possible complication of the SAH accompanied with intracranial subdural hematoma.

10. **Answer:** True

Options: A) True; B) False

Note: Transgastric endoscopic splenectomy in a porcine model appears technically feasible. Additional long-term survival experiments are planned.

Long Form Response

11. **Answer:** Point mutations are repaired, introns are removed, and both a 5 cap and a poly-A tail are added after the preRNA leaves the nucleus.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** The filamentous appendages are not flagella, but pili (or fimbriae).. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.